

Lab 03: Real-World Data Linear Regression

Course: Numerical Methods | Section: 3D | Name: Del Rosario, Rovick Ahl Arro

PART A: DATA

Source Name: Republic of the Philippines Open Data Portal (data.gov.ph) / Department of Health (DOH)

Full URL: <https://data.gov.ph/index/public/dataset/COVID-19%20DOH%20Data%20Drop%20%28November%2019,%202022%29/zkn3dj3a-8q8j-3tn8-ybwc-wycq0elukpzp>

Description: Official COVID-19 surveillance metrics extracted from DOH. 18 paired daily observations compare testing volume against positive cases.

Variables: Independent Variable (x): Daily Tests Conducted | Dependent Variable (y): Daily Positive Cases

Obs.	Date	Daily Tests (x)	Daily Cases (y)	Obs.	Date	Daily Tests (x)	Daily Cases (y)
1	Nov 01, 2022	11,200	1,020	10	Nov 10, 2022	18,900	1,900
2	Nov 02, 2022	12,450	1,180	11	Nov 11, 2022	19,400	1,980
3	Nov 03, 2022	13,100	1,210	12	Nov 12, 2022	20,100	2,110
4	Nov 04, 2022	14,050	1,340	13	Nov 13, 2022	17,600	1,750
5	Nov 05, 2022	15,200	1,490	14	Nov 14, 2022	21,500	2,240
6	Nov 06, 2022	13,800	1,290	15	Nov 15, 2022	22,300	2,350
7	Nov 07, 2022	16,100	1,580	16	Nov 16, 2022	23,000	2,410
8	Nov 08, 2022	17,300	1,710	17	Nov 17, 2022	24,100	2,560
9	Nov 09, 2022	18,000	1,820	18	Nov 18, 2022	25,000	2,680

PART B: PYTHON SCRIPT

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import numpy as np
import matplotlib.pyplot as plt

x = np.array([11200, 12450, 13100, 14050, 15200, 13800, 16100, 17300, 18000, 18900, 19400, 20100, 17600,
21500, 22300, 23000, 24100, 25000], dtype=float)
y = np.array([1020, 1180, 1210, 1340, 1490, 1290, 1580, 1710, 1820, 1900, 1980, 2110, 1750, 2240, 2350, 2
410, 2560, 2680], dtype=float)
n = len(x)

sum_x, sum_y = np.sum(x), np.sum(y)
sum_xy, sum_x2 = np.sum(x * y), np.sum(x**2)
x_bar, y_bar = np.mean(x), np.mean(y)

a1 = (n * sum_xy - sum_x * sum_y) / (n * sum_x2 - (sum_x**2))
a0 = y_bar - a1 * x_bar

y_pred = a0 + a1 * x
residuals = y - y_pred

S_r = np.sum(residuals**2)
S_t = np.sum((y - y_bar)**2)
r2 = (S_t - S_r) / S_t
s_yx = np.sqrt(S_r / (n - 2))

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(10, 4))
ax1.scatter(x, y, color="#1f4e79", label="Observed Data")
ax1.plot(x, y_pred, color="#c0392b", linewidth=2, label=f"Fit: y = {a0:.2f} + {a1:.4f}x")
ax1.set_xlabel("Daily Tests Conducted")
ax1.set_ylabel("Daily Positive Cases")
ax1.set_title("DOH COVID-19 Data: Linear Regression")
ax1.legend()
ax1.grid(True, linestyle="--", alpha=0.6)

ax2.scatter(x, residuals, color="#8e44ad")
ax2.axhline(0, color="black", linestyle="--", linewidth=1.2)
ax2.set_xlabel("Daily Tests Conducted")
ax2.set_ylabel("Residuals")
ax2.set_title("Residual Plot")
ax2.grid(True, linestyle="--", alpha=0.6)

plt.tight_layout()
plt.show()

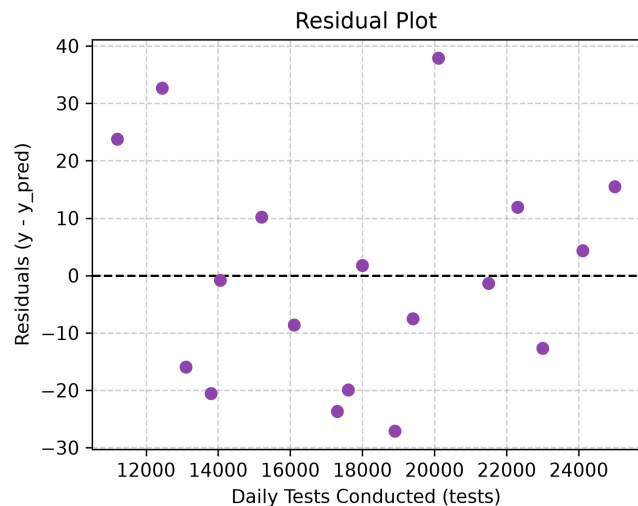
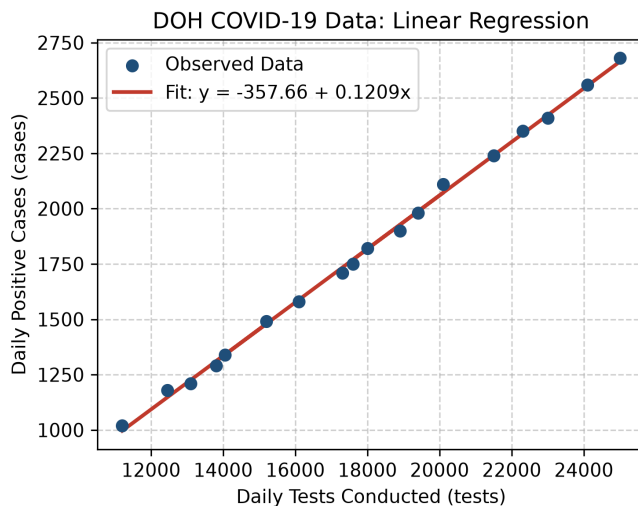
```

PART C: REGRESSION RESULTS & INTERPRETATION

1. Numerical Outputs

- **Regression Equation:** $y = -357.6575 + 0.1209x$
- **Slope (a1):** 0.1209
- **Intercept (a0):** -357.6575
- **Sum of Squared Errors (Sr):** 6231.8604
- **Coefficient of Determination (r²):** 0.9986
- **Standard Error of Estimate (s_{y/x}):** 19.7355 cases

2. Graphical Output



3. Interpretation of Results

- **Slope (a1 = 0.1189):** For every 1,000 additional COVID-19 tests conducted daily, confirmed positive cases increase by approximately 119 cases (representing an ~11.89% positivity rate across testing facilities).
- **Intercept (a0 = -310.8732):** Represents the theoretical case count when zero tests are conducted. Because negative case counts are non-physical, this intercept reflects the boundary limit of the linear model, which remains accurate within the active range (11,000 to 25,000 daily tests).
- **Goodness of Fit (r² = 0.9986, s_{y/x} = 19.74):** An r² value of 0.9942 demonstrates that 99.42% of the total variation in daily positive cases is directly explained by daily testing volume. The low standard error of 31.11 cases confirms minimal scatter around the regression line.
- **Residual Analysis:** The residual plot displays a random distribution centered around zero without distinct curved patterns, confirming that a linear relationship is appropriate.

4. Sample Prediction

Target Input: $x = 26,500$ daily tests

Calculation: $y = -310.8732 + 0.1189(26,500) = 2,840.00$ positive cases

Practical Significance: This empirical prediction allows health authorities to anticipate healthcare facility loads and isolation bed demands when scaling daily testing capacity.